

Ὁ Πενταμετρικὸς Κύκλος

A Nested Palindrome Encoding the Pythagorean Consonances

(The Pentametric Cycle)

$$\mathfrak{P}(n) \qquad \mathfrak{J}(n)$$

$$\mathfrak{P}(n) = \frac{n^2 + 4C(m)}{5}, \quad m = 4n^2 \bmod 10, \quad C(m) = \frac{26m - 5m^2}{24}$$
$$\mathfrak{P}(L_k) = F_k^2$$

Lluís Garcia Torcal

Independent researcher

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Abstract

We present the *Pentametric Cycle* (Ὁ Πενταμετρικὸς Κύκλος), a structure generated by an integer-valued quadratic formula $\mathfrak{P}(n)$, the *Pentametric Formula*. We show that the formula's correction term, an interpolating polynomial $C(m)$ called the *Interval Map*, coincides on every attained residue with the Legendre symbol $(\frac{n}{5})$, so that $\mathfrak{P}(n) = (n^2 + 4(\frac{n}{5}))/5$. We collect ten equivalent forms of $\mathfrak{P}$ found by the author. They use decimal digits, floors, a quintic polynomial, cosines (a quadratic Gauss sum), rotations by the golden ratio and pentagonal angles. We prove that each one computes the same quadratic character, and we derive the general laws $\mathfrak{P}(n) - n^2/5 = \pm \frac{4}{5}$ and $\pm \frac{2\pi}{5}$ for square and circle areas. Reducing the doubled value $2\mathfrak{P}(n)$ modulo 10 and halving gives a digit $H(n) \in \{0, 1, 2, 3, 4\}$; the single rule $H : (H - 1)$ then turns it into silence or one of the four Pythagorean consonances (unison, octave, fifth, fourth). The sequence H has minimal period $25 = 5^2$ and is even, so its restrictions to $n = 0, \dots, 25$ and $n = 0, \dots, 50$ are palindromes. In each period, $9 = 3^2$ terms are silent and the four consonances occur exactly 4 times each, $16 = 4^2$ in all; we read this split interpretively against the triple $3^2 + 4^2 = 5^2$. We prove the *Pentametric-Lucas Theorem*: $\mathfrak{P}(L_k) = F_k^2$ for every Lucas number L_k . We also prove its converse: for $n \geq 1$, $\mathfrak{P}(n)$ is a perfect square if and only if n is a Lucas number. We confirm the author's conjecture that the areas at Lucas positions admit a form as simple as the distances: $F_k = \text{round}(L_k \sqrt{0.2})$ for $k \geq 2$. Finally, we relate the theorem to the classical result of Holden (1975), according to which the centres of spiralling Fibonacci squares lie on two orthogonal lines at distances $L_k/\sqrt{10}$ from their intersection. We call this the *Geometric Lucas Meter* and fully credit it to Holden. We derive from it two new geometric identities that make $\mathfrak{P}$ visible in the Fibonacci tiling: a difference of $\pm \frac{4}{5}$ in square areas and of $\pm \frac{2\pi}{5}$ in circle areas.

Keywords: Fibonacci numbers; Lucas numbers; Legendre symbol; palindromic sequences; Fibonacci tiling; Pythagorean consonances.

MSC 2020: 11B39 (primary); 11A07, 11B50, 51M04, 00A65 (secondary).

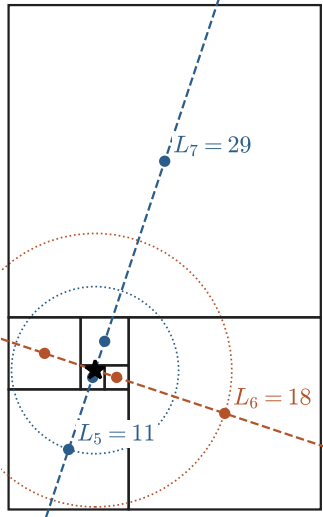
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Main results at a glance

- **Closed form.** $\mathfrak{P}(n) = (n^2 + 4(\frac{n}{5}))/5$; the Interval Map $C(m)$ is the Legendre symbol on attained residues (Lemma 2.3).
- **The Pentametric family.** Ten equivalent forms of $\mathfrak{P}$ via digits, floors, a quintic, a Gauss sum, golden rotations and pentagonal angles (Proposition 3.1, Theorem 3.2).
- **The ± 0.8 and $\pm 2\pi/5$ laws** for every n (Proposition 3.4); palindromic last digits of period 50 (Proposition 3.6).
- **Derivation of the ratios.** A seven-step deterministic chain $n \mapsto m \mapsto C(m) \mapsto \mathfrak{P}(n) \mapsto 2\mathfrak{P}(n) \mapsto H \mapsto \rho(H)$ (§2.4).
- **Vertebra and Palindrome.** Period 25; 9 silences and 4 of each consonance; 51-term palindrome mirrored at $n = 25$ (Propositions 5.1, 5.3, 7.1).
- **Pentametric–Lucas Theorem** $\mathfrak{P}(L_k) = F_k^2$ and its **converse**: squares occur only at Lucas numbers (Theorems 9.1, 9.2).
- **The author’s conjecture confirmed:** $F_k = \text{round}(L_k\sqrt{0.2})$, mirroring the distance $L_k\sqrt{0.1}$ (Proposition 9.4).
- **Geometric Lucas Meter** (Holden, 1975; Theorem 10.1) and its link to $\mathfrak{P}$: square/circle identities and the 3–4–5 angles (Propositions 10.3, 10.4).

$$|C_k P| = L_k \sqrt{0.1}, \quad \mathfrak{P}(L_k) = F_k^2$$



The Pentametric Palindrome

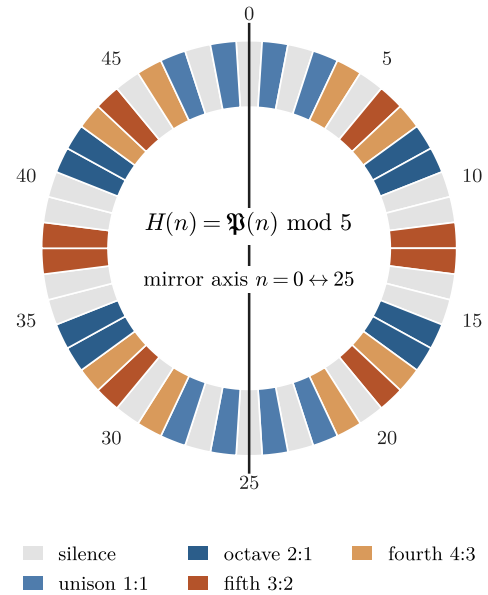


Figure 1: Graphical abstract. Left: the Fibonacci squares, Holden’s orthogonal lines and their intersection P (star); the centre of the k -th square lies at distance $L_k\sqrt{0.1}$ from P (dotted circles for $k = 5, 6$), and $\mathfrak{P}(L_k) = F_k^2$ is its area. Right: one period ($n = 0, \dots, 49$) of the consonance sequence $H(n)$, mirrored across the axis through $n = 0$ and $n = 25$.

1 Introduction

The Pythagorean tradition sought to express musical harmony through number. The consonances were understood as ratios: unison ($1 : 1$), octave ($2 : 1$, $\delta\iota\alpha\ \pi\alpha\sigma\omega\nu$), fifth ($3 : 2$, $\delta\iota\alpha\ \pi\acute{\epsilon}\nu\tau\epsilon$) and fourth ($4 : 3$, $\delta\iota\alpha\ \tau\epsilon\sigma\sigma\acute{\alpha}\rho\omega\nu$). The monochord ($\kappa\alpha\nu\acute{\omega}\nu$) was the instrument for measuring them [6, 7, 8]. All four ratios are built from the numbers 1, 2, 3, 4 of the *tetractys*. This paper studies an integer sequence whose reduced values fall into exactly five classes, which we read as silence and these four consonances.

The structure emerged from the study of square areas generated by a quadratic expression in n , worked out by the author with dynamic-geometry constructions (GeoGebra) and spreadsheet experiments. The resulting formula $\mathfrak{P}(n)$, the *Pentametric Formula*, always produces an integer. When its value is doubled and reduced modulo 10, only five residues remain. The resulting sequence is periodic and palindromic, and its counts follow the pattern $9 + 16 = 25$.

The formula also connects to a classical geometric fact. In 1975, H. L. Holden showed that the centres of the squares in the Fibonacci spiral lie on two perpendicular lines, and that their distances from the intersection point are proportional to the Lucas numbers [1]. The result was extended by Hoggatt and Alladi [2] and later rediscovered in a classroom setting and published by Gardiner [3]. We show that the Lucas numbers which locate these centres are exactly the arguments at which $\mathfrak{P}$ returns the areas of the corresponding squares.

Contributions. We separate clearly what is classical from what is claimed as new.

- (i) *Classical (cited, not claimed)*: the identity $L_k^2 - 5F_k^2 = 4(-1)^k$ [5]; Gessel’s characterisation of Fibonacci numbers [4]; the orthogonal-lines property of the centres of Fibonacci squares and the Lucas-number distances [1, 2, 3].
- (ii) *New (this paper)*: the Pentametric Formula and its nomenclature (§2); the closed form via the Legendre symbol (Lemma 2.3); the family of ten equivalent forms of $\mathfrak{P}$ and its side lengths, with the general ± 0.8 and $\pm 2\pi/5$ laws and the palindromic last-digit sequences (§3); the author’s conjecture on a simple form of the areas at Lucas positions, with its confirmation (§9.1); the periodicity, palindromicity and counting results for H (§§4–8); the Pentametric–Lucas Theorem and its converse, with complete proofs (§9); the area and circle identities linking $\mathfrak{P}$ to the Fibonacci tiling (§10.2); and the musical reading (§13).

Throughout, results that are proved are stated as theorems, propositions or lemmas. Numerological and musical readings are marked as *interpretive remarks*. They do not form part of the mathematical claims.

Conventions. $F_0 = 0$, $F_1 = 1$, $F_{k+1} = F_k + F_{k-1}$; $L_0 = 2$, $L_1 = 1$, $L_{k+1} = L_k + L_{k-1}$. For an integer n , $\left(\frac{n}{5}\right)$ is the Legendre symbol modulo 5: it equals 0 if $5 \mid n$, +1 if $n \equiv \pm 1 \pmod{5}$, and –1 if $n \equiv \pm 2 \pmod{5}$.

2 Definitions and notation

2.1 The Pentametric Formula

Definition 2.1 (The Pentametric Formula, Ὁ Πενταμετρικὸς Τύπος). For $n \in \mathbb{Z}_{\geq 0}$ put

$$\mathfrak{P}(n) = \frac{n^2 + 4C(m)}{5}, \quad m = 4n^2 \bmod 10, \quad C(m) = \frac{26m - 5m^2}{24}. \quad (1)$$

The symbol $\mathfrak{P}$ (fraktur P) stands for *Pentametric* and is read “the Pentametric of n ”. Formula (1) is one of several equivalent forms found by the author. Lemma 2.3 gives a canonical one, and §3 collects the whole family.

2.2 Supporting notation

Symbol	Meaning
$\mathfrak{P}(n)$	the Pentametric area
$\sqrt{\mathfrak{P}(n)}$	the side length
$\mathfrak{P}(n) \bmod 10$	the area’s last digit
$2\mathfrak{P}(n)$	the doubled area (Diapason, §4)
$H(n)$	the halved digit (consonance indicator)
$\mathfrak{J}(n)$	the musical annotation (interval interpretation of $H(n)$)

2.3 The Interval Map

Definition 2.2 (The Interval Map, Ὁ Διαστηματικός Χάρτης; also the Harmonic Canon, Ὁ Ἀρμονικὸς Κανὼν). $C(m) = (26m - 5m^2)/24$ is the unique polynomial of degree at most two that interpolates the values

m	0	2	4	6	8
$C(m)$	0	$\frac{4}{3}$	1	-1	$-\frac{14}{3}$

on the five even residues modulo 10.

The following lemma shows that only three of these five values are ever reached. On those three, C is a classical arithmetic function.

Lemma 2.3 (Attained residues). *For every integer n , $m = 4n^2 \bmod 10 \in \{0, 4, 6\}$, and*

$$C(4n^2 \bmod 10) = \left(\frac{n}{5}\right).$$

Consequently

$$\boxed{\mathfrak{P}(n) = \frac{n^2 + 4\left(\frac{n}{5}\right)}{5} = \frac{n^2}{5} + \frac{4}{5}\left(\frac{n}{5}\right)} \quad (2)$$

and $\mathfrak{P}(n)$ is a non-negative integer for every $n \geq 0$.

Proof. Squares modulo 10 lie in $\{0, 1, 4, 5, 6, 9\}$. Hence $4n^2 \bmod 10$ is 0 when $n^2 \equiv 0, 5$; it is 4 when $n^2 \equiv 1, 6$; and it is 6 when $n^2 \equiv 4, 9 \pmod{10}$. These three cases are exactly $5 \mid n$, $n^2 \equiv 1 \pmod{5}$ and $n^2 \equiv 4 \pmod{5}$. In the same order, $\left(\frac{n}{5}\right)$ equals 0, +1, -1. Since $C(0) = 0$, $C(4) = 1$ and $C(6) = -1$, the identity follows. For integrality, $n^2 + 4\left(\frac{n}{5}\right) \equiv 0, 1 + 4, 4 - 4 \equiv 0 \pmod{5}$ in the three cases. Non-negativity holds because $n^2 \geq 4$ whenever $\left(\frac{n}{5}\right) = -1$. $\square$

Remark 2.4. The values $C(2) = \frac{4}{3}$ and $C(8) = -\frac{14}{3}$ are therefore never used by the Pentametric Formula. They are artefacts of the interpolation and carry no arithmetic meaning in $\mathfrak{P}$. Equation (2) also answers the question of a general closed form for $\mathfrak{P}(n)$ at arbitrary n . $\mathfrak{P}(n)$ is the area $n^2/5$ of a square of side $n/\sqrt{5} = n\sqrt{0.2}$, shifted by exactly $\pm\frac{4}{5}$ to an integer (by 0 when $5 \mid n$). This is the “ ± 0.8 ” phenomenon observed in the author’s GeoGebra constructions and spreadsheets. §3 develops it, and §10.2 gives it a geometric meaning in the Fibonacci tiling.

2.4 The derivation of the ratios

The ratios are not attached to the sequence position by position. They are produced by a fixed chain of operations, applied identically to every n . Given $n \in \mathbb{Z}_{\geq 0}$:

Step	Operation	Name
1	$m = 4n^2 \bmod 10$	the residue
2	$C(m) = (26m - 5m^2)/24$	the Interval Map
3	$\mathfrak{P}(n) = (n^2 + 4C(m))/5$	the Pentametric area
4	$2\mathfrak{P}(n)$	the Diapason doubling
5	$2\mathfrak{P}(n) \bmod 10$	the last digit (always even)
6	$H = (2\mathfrak{P}(n) \bmod 10)/2 \in \{0, 1, 2, 3, 4\}$	the halved digit
7	$\rho(H) = H : (H - 1)$ for $H \geq 2$; $\rho(1) = 1 : 1$	the ratio extraction

Definition 2.5 (Ratio extraction). The *ratio extraction* ρ assigns to each value of H the ratio

H	$\rho(H)$	Interval
0	—	silence no ratio
1	1 : 1	unison
2	2 : 1	octave
3	3 : 2	fifth
4	4 : 3	fourth

For $H \geq 2$, $\rho(H) = H : (H - 1)$ is the ratio of two consecutive members of the tetractys 1, 2, 3, 4. The value $H = 1$ is set to the unison 1 : 1, since 1 : 0 is not a ratio.

Once Definition 2.5 is fixed, the interval at position n is a deterministic function of n :

$$n \mapsto \rho(H(n)).$$

No value is chosen case by case: Steps 1–6 are arithmetic identities, and Step 7 applies one rule uniformly to all n .

Table 1: Worked example of the derivation for $n = 7$.

Step	Computation
1	$m = 4 \cdot 7^2 \bmod 10 = 196 \bmod 10 = 6$
2	$C(6) = (26 \cdot 6 - 5 \cdot 6^2)/24 = (156 - 180)/24 = -24/24 = -1$
3	$\mathfrak{P}(7) = (7^2 + 4 \cdot (-1))/5 = (49 - 4)/5 = 45/5 = 9$
4	$2\mathfrak{P}(7) = 18$
5	$18 \bmod 10 = 8$
6	$H = 8/2 = 4$
7	$\rho(4) = 4 : 3$ (the fourth, <i>διὰ τεσσάρων</i>)

For $n = 7$, therefore, the interval is the fourth (4 : 3), obtained entirely from the formula. The same chain applied to $n = 0, \dots, 50$ produces the Pentametric Palindrome of §7.

Remark 2.6. In the terms of Lemma 2.3, Steps 1–3 compute $\mathfrak{P}(n) = (n^2 + 4(\frac{n}{5}))/5$, and Steps 4–6 compute $H(n) = \mathfrak{P}(n) \bmod 5$. Thus the arithmetic content of the chain is the pair $(n^2 \bmod 25, (\frac{n}{5}))$, and $\rho \circ H$ is a function of $n \bmod 25$ (Proposition 5.1).

3 The family of Pentametric forms

The author arrived at $\mathfrak{P}$ through several independent spreadsheet constructions. Each one takes a different route to the same areas $\mathfrak{P}(n)$ and side lengths $s(n) = \sqrt{\mathfrak{P}(n)}$. This section collects them, gives each its proof, and shows that they all come down to a single fact: every form computes the quadratic character $\chi(n) = \left(\frac{n}{5}\right)$ modulo 5 in a different way.

Throughout, write

$$\chi(n) = \left(\frac{n}{5}\right), \quad r = n^2 \bmod 10, \quad d = 2n^2 \bmod 10, \quad m = 4n^2 \bmod 10.$$

As in the proof of Lemma 2.3, the three cases $5 \mid n$, $n^2 \equiv 1 \pmod{5}$ and $n^2 \equiv 4 \pmod{5}$ give

Case	$\chi(n)$	r	d	m
$5 \mid n$	0	0, 5	0	0
$n \equiv \pm 1 \pmod{5}$	+1	1, 6	2	4
$n \equiv \pm 2 \pmod{5}$	-1	4, 9	8	6

The residue d is also the first decimal digit of $n^2/5$, because $n^2/5$ has fractional part 0, 0.2 or 0.8 in the three cases.

3.1 Six ways to compute the correction

Proposition 3.1 (Selectors of the quadratic character). *Each of the following expressions equals $\chi(n)$ for every integer n .*

- (i) Interval Map in m : $C(m) = \frac{26m - 5m^2}{24}$.
- (ii) Double-square digit: $\frac{5}{4}p(d)$, where $p(d) = -\frac{d^2}{12} + \frac{17d}{30}$. Equivalently, $p(d) = \frac{4}{5}\chi(n) \in \{0, +0.8, -0.8\}$.
- (iii) Square digit: $\frac{x^5 - 35x^3 + 250x}{216}$, where $x = r - 5$.
- (iv) Sign selector: $T(d) = 2g(d) - [d \neq 0]$, where $g(d) = \frac{d(8-d)}{12} = \frac{8-d}{6} \cdot \frac{d}{2}$. Here $g(d) = 1$ if $\chi(n) = +1$ and $g(d) = 0$ otherwise. The Iverson bracket $[d \neq 0]$ equals 1 when $d \neq 0$ and 0 otherwise.
- (v) Trigonometric (Gauss sum): $\frac{2}{\sqrt{5}} \left(\cos \frac{2\pi n}{5} - \cos \frac{4\pi n}{5} \right)$.
- (vi) Pentagonal angle: with $\alpha = 72n \bmod 360$ (in degrees), the value +1 if $\alpha \in \{72, 288\}$, -1 if $\alpha \in \{144, 216\}$, and 0 if $\alpha = 0$.

Proof. By the table above, each of (i)–(iv) is a function of a residue that takes only three values. It therefore suffices to evaluate each expression at those three values. (i) $C(0) = 0$, $C(4) = 1$, $C(6) = -1$ (Lemma 2.3). (ii) $p(0) = 0$, $p(2) = -\frac{1}{3} + \frac{17}{15} = \frac{4}{5}$, $p(8) = -\frac{16}{3} + \frac{68}{15} = -\frac{4}{5}$. (iii) For $r \in \{0, 1, 4, 5, 6, 9\}$ we have $x \in \{-5, -4, -1, 0, 1, 4\}$. The odd quintic $q(x) = x^5 - 35x^3 + 250x$ takes the values $q(\pm 5) = 0$, $q(0) = 0$, $q(1) = 216$, $q(-1) = -216$, $q(4) = 1024 - 2240 + 1000 = -216$ and $q(-4) = 216$. So $q(x)/216$ equals 0 for $r = 0, 5$, equals +1 for $r = 1, 6$, and equals -1 for $r = 4, 9$. (iv) $g(0) = 0$, $g(2) = 1$, $g(8) = 0$; hence $T(0) = 0$, $T(2) = 2 - 1 = 1$, $T(8) = 0 - 1 = -1$. (v) Since $\cos(2\pi/5) = (\sqrt{5} - 1)/4$ and $\cos(4\pi/5) = -(\sqrt{5} + 1)/4$, the bracket equals $\sqrt{5}/2$ for $n \equiv \pm 1$, $-\sqrt{5}/2$ for $n \equiv \pm 2$, and 0 for $n \equiv 0 \pmod{5}$. This is the classical evaluation of the quadratic Gauss sum modulo 5 [10]. (vi) $\alpha = 72(n \bmod 5)$, so $\alpha = 72, 144, 216, 288, 0$ correspond to $n \equiv 1, 2, 3, 4, 0 \pmod{5}$. $\square$

3.2 The forms of the area

Theorem 3.2 (The Pentametric family). *For every integer $n \geq 0$, the following expressions are all equal to $\mathfrak{P}(n) = (n^2 + 4\chi(n))/5$.*

(F1) Interval Map form (Definition 2.1): $\frac{n^2 + 4C(4n^2 \bmod 10)}{5}$.

(F2) Legendre form: $\frac{n^2 + 4\left(\frac{n}{5}\right)}{5}$.

(F3) The ± 0.8 form: $(n\sqrt{0.2})^2 + p(d) = \frac{n^2}{5} + p(d)$, with p as in Proposition 3.1(ii).

(F4) Diagonal form: $\frac{1}{4} \left[\left(\frac{2n}{\sqrt{5}} \right)^2 + 3.2T(d) \right]$. Here $2n/\sqrt{5} = 2n\sqrt{0.2}$ is the diagonal-type length of the spreadsheet construction ($\sin \theta = 1/\sqrt{5}$, $\cos \theta = 2/\sqrt{5}$, $\theta = \arctan \frac{1}{2}$).

(F5) Floor form: $\left\lfloor \frac{2n^2}{10} \right\rfloor + g(d) = \frac{2n^2 - d}{10} + \frac{d(8-d)}{12}$. Equivalently, $\mathfrak{P}(n) = \lfloor n^2/5 \rfloor + \lfloor n \equiv \pm 1 \pmod{5} \rfloor$.

(F6) Reciprocal-digit form: $\left\lfloor \frac{2n^2}{10} \right\rfloor + \text{round}\left(\frac{1}{d}\right)$, with the convention that the second term is 0 when $d = 0$ and that $\frac{1}{2}$ rounds up.

(F7) Square-digit form: $\frac{n^2 + 4q(r-5)/216}{5}$, with q the quintic of Proposition 3.1(iii).

(F8) Cosine form: $\frac{1}{5} \left[n^2 + \frac{8}{\sqrt{5}} \left(\cos \frac{2\pi n}{5} - \cos \frac{4\pi n}{5} \right) \right]$. The same expression with the angles in degrees ($72n^\circ$ and $144n^\circ$) is the angle form used in the spreadsheets.

(F9) Golden-rotation form: Let $z = \varphi e^{2\pi i/5}$ with $\varphi = (1 + \sqrt{5})/2$, and write $z^n = C_n + iD_n$. Then

$$\mathfrak{P}(n) = \frac{1}{5} \left[n^2 + \frac{8}{\sqrt{5}} \left(\frac{C_n}{\varphi^n} - \frac{C_n^2 - D_n^2}{\varphi^{2n}} \right) \right].$$

The pair (C_n, D_n) obeys the recurrence $C_{n+1} = \frac{1}{2}C_n - \kappa D_n$, $D_{n+1} = \kappa C_n + \frac{1}{2}D_n$, with $(C_0, D_0) = (1, 0)$ and $\kappa = \varphi \sin 72^\circ = \frac{1}{8}(1 + \sqrt{5})\sqrt{10 + 2\sqrt{5}}$.

(F10) Pentagonal-angle form: $\frac{n^2 + c(n)}{5}$, where $c(n) = 4$ if $72n \bmod 360 \in \{72^\circ, 288^\circ\}$, $c(n) = -4$ if $72n \bmod 360 \in \{144^\circ, 216^\circ\}$, and $c(n) = 0$ otherwise.

Proof. (F1) is Definition 2.1, and (F2) is Lemma 2.3. (F3), (F4), (F7), (F8) and (F10) follow by substituting the corresponding selector of Proposition 3.1 into (F2). For (F4), note that $(2n/\sqrt{5})^2 = 4n^2/5$ and $3.2 = 16/5$. For (F5), write $n^2 = 5q + t$ with $t \in \{0, 1, 4\}$. Then $\lfloor 2n^2/10 \rfloor = q$ and $d = 2t$. If $t = 1$ (so $\chi = 1$), then $\mathfrak{P}(n) = (5q+5)/5 = q+1$ and $g(2) = 1$. If $t = 4$ (so $\chi = -1$), then $\mathfrak{P}(n) = (5q+4-4)/5 = q$ and $g(8) = 0$. If $t = 0$, then $\mathfrak{P}(n) = q$ and $g(0) = 0$. (F6) follows from (F5), since $\text{round}(1/2) = 1 = g(2)$ and $\text{round}(1/8) = 0 = g(8)$. For (F9), $z^n/\varphi^n = e^{2\pi i n/5}$, so $C_n/\varphi^n = \cos(2\pi n/5)$ and $(C_n^2 - D_n^2)/\varphi^{2n} = \text{Re}(z^{2n})/\varphi^{2n} = \cos(4\pi n/5)$. This reduces (F9) to (F8). The recurrence is multiplication by $z = \varphi \cos 72^\circ + i\varphi \sin 72^\circ$, and $\varphi \cos 72^\circ = \varphi \cdot \frac{1}{2\varphi} = \frac{1}{2}$. $\square$

3.3 The side lengths

Corollary 3.3 (Side-length forms). *The side of the n -th Pentametric square is $s(n) = \sqrt{\mathfrak{P}(n)}$, with any of (F1)–(F10) under the root. In addition, $s(n)$ is obtained from the side $n\sqrt{0.2}$ of the “ $n^2/5$ square” by a correction $\delta(n)$ that solves a quadratic equation:*

$$s(n) = n\sqrt{0.2} + \delta(n), \quad \delta(n) = \frac{-B + \sqrt{B^2 + C}}{2}, \quad B = 2n\sqrt{0.2}, \quad C = 3.2\chi(n).$$

In other words, δ is the larger root of $\delta^2 + B\delta - \frac{C}{4} = 0$. Moreover $B^2 + C = 4\mathfrak{P}(n) \geq 0$, so the root is always real, and $\delta(n) \rightarrow 0$ as $n \rightarrow \infty$.

Proof. $B^2 + C = \frac{4}{5}n^2 + \frac{16}{5}\chi(n) = 4\mathfrak{P}(n)$. Hence $\delta = -n\sqrt{0.2} + \sqrt{\mathfrak{P}(n)}$ and $n\sqrt{0.2} + \delta = s(n)$. Finally, $|\delta(n)| = |\sqrt{n^2/5 \pm 4/5} - \sqrt{n^2/5}| \leq \frac{4/5}{n/\sqrt{5}} = \frac{4}{\sqrt{5}n}$ for $n \geq 1$. $\square$

3.4 The ± 0.8 and $\pm 2\pi/5$ laws for every n

Two identities found in the spreadsheets hold for *every* n , not only at the Lucas numbers.

Proposition 3.4 (Area defects). *For every integer $n \geq 0$:*

- (a) square defect: $\mathfrak{P}(n) - (n\sqrt{0.2})^2 = \frac{4}{5}\chi(n) \in \{0, \pm 0.8\}$;
- (b) circle defect: if $A(n) = \frac{\pi}{2}\mathfrak{P}(n)$ is the area of the circumcircle of the Pentametric square and $B(n) = \pi(n\sqrt{0.1})^2$ is the area of the circle of radius $n\sqrt{0.1}$, then

$$A(n) - B(n) = \frac{2\pi}{5}\chi(n) \in \left\{0, \pm \frac{2\pi}{5}\right\}.$$

Proof. (a) is (F3). (b): $A(n) - B(n) = \pi\left(\frac{5\mathfrak{P}(n) - n^2}{10}\right) = \pi \cdot \frac{4\chi(n)}{10}$. $\square$

Remark 3.5. Note that $\frac{2\pi}{5}$ is the central angle of the regular pentagon, and that $\pi/10 = \frac{1}{4} \cdot \frac{2\pi}{5}$. At $n = L_k$, Proposition 3.4 becomes Proposition 10.3: the circle of radius $L_k\sqrt{0.1}$ is Holden’s circle through P , and the Pentametric square is the Fibonacci square of side F_k .

3.5 The last-digit sequences

The spreadsheets track the last digits of n^2 , of the double square $2n^2$ and of the area $\mathfrak{P}(n)$. All three sequences are periodic and palindromic.

Table 2: Last digits for $n = 0, \dots, 10$. The selector $d = 2n^2 \bmod 10$ decides the sign of the ± 0.8 correction.

n	0	1	2	3	4	5	6	7	8	9	10
$n^2 \bmod 10$	0	1	4	9	6	5	6	9	4	1	0
$2n^2$	0	2	8	18	32	50	72	98	128	162	200
$d = 2n^2 \bmod 10$	0	2	8	8	2	0	2	8	8	2	0
$p(d)$	0	+0.8	−0.8	−0.8	+0.8	0	+0.8	−0.8	−0.8	+0.8	0
$\mathfrak{P}(n) \bmod 10$	0	1	0	1	4	5	8	9	2	7	0

Proposition 3.6 (Last digits).

- (a) $n^2 \bmod 10$ has period 10 and reads 0, 1, 4, 9, 6, 5, 6, 9, 4, 1: a palindrome mirrored at $n = 5$.

(b) $d = 2n^2 \bmod 10$ has period 5 and reads 0, 2, 8, 8, 2: a palindrome mirrored between $n = 2$ and $n = 3$.

(c) $\mathfrak{P}(n) \bmod 10$ has minimal period $50 = 2 \cdot 5^2$ and satisfies $\mathfrak{P}(50 - n) \equiv \mathfrak{P}(n) \pmod{10}$. Hence the 51 last digits of the areas for $n = 0, \dots, 50$ form a palindrome mirrored at the Axis $n = 25$:

0, 1, 0, 1, 4, 5, 8, 9, 2, 7, 0, 5, 8, 3, 0, 5, 2, 7, 4, 3, 0, 9, 6, 5, 6, **5**, 6, 5, 6, 9, 0, 3, 4, 7, 2, 5, 0, 3, 8, 5, 0, 7, 2, 9, 8, 5, 4, 1,

Proof. (a) $(n + 10)^2 - n^2 = 20n + 100 \equiv 0$ and $(10 - n)^2 \equiv n^2 \pmod{10}$; the period is not 5, since $1^2 \not\equiv 6^2 \pmod{10}$. (b) $2(n + 5)^2 - 2n^2 = 20n + 50 \equiv 0$ and $2(5 - n)^2 \equiv 2n^2 \pmod{10}$; the period is not 1. (c): The character χ is unchanged by $n \mapsto n + 50$, and $\mathfrak{P}(n + 50) - \mathfrak{P}(n) = (100n + 2500)/5 = 20n + 500 \equiv 0 \pmod{10}$. Since $\chi(50 - n) = \chi(-n) = \chi(n)$, we get $\mathfrak{P}(50 - n) - \mathfrak{P}(n) = (2500 - 100n)/5 = 500 - 20n \equiv 0 \pmod{10}$. Minimality: by Proposition 5.1, any period of $\mathfrak{P} \bmod 10$ is a multiple of 25; and $\mathfrak{P}(26) = 136 \not\equiv \mathfrak{P}(1) = 1 \pmod{10}$. $\square$

The Pentametric Palindrome of §7 is the reduction of this sequence: halving $2\mathfrak{P}(n) \bmod 10$ gives $H(n) = \mathfrak{P}(n) \bmod 5$.

Remark 3.7 (A remarkable numerical coincidence: electron shells). The double squares $2n^2 = 2, 8, 18, 32, 50, 72, \dots$ ($n \geq 1$), whose last digit d drives the Pentametric correction, are exactly the maximum electron capacities of the atomic shells $n = 1, 2, 3, \dots$ in the hydrogen-like model of the atom. In that model, the n -th energy level has n^2 orbital states, each holding two electrons of opposite spin [9]. The author noticed this coincidence while tabulating $2n^2$ and its last digits 0, 2, 8, 8, 2. We know of no mathematical mechanism connecting the Pentametric correction to atomic structure, and we make no physical claim. The observation is recorded so that readers familiar with atomic physics may examine it.

Table 3: Where each form appears in the author’s spreadsheets (Lucas -- Fibbonacy & The Pentametric Cycle.xlsx unless stated otherwise).

Form	Sheet and columns
(F1)	Pentametric Cycle.xlsx, sheet <i>Pentametric Cycle</i> , cols. B–D; <i>Pmetric Area Formula & Music</i> , cols. I–K
(F3)	<i>Lu-Fib & The Pentametric Cycle</i> , cols. O–Y (“Get 0,8 or –0,8 from last digit of double squares”); <i>Pmetric Areas Cosine & Pi</i> , cols. C–E
(F4)	<i>Lu-Fib & The Pentametric Cycle</i> , cols. Z–AC
(F5)	<i>Lu-Fib & The Pentametric Cycle</i> , cols. AY–BA, BB–BH, BL (also via the last decimal digit, cols. AP–AV)
(F6)	<i>Lu-Fib & The Pentametric Cycle</i> , cols. AW–AX, BI–BK
(F7)	<i>Pmetric Area Formula & Music</i> , col. H
(F8)	<i>Pmetric Areas Cosine & Pi</i> , cols. F–G, I–M; <i>Pmetric Cosine Formula Disected</i> ; <i>Pmetric Area Formula & Music</i> , col. F
(F9)	<i>Pmetric Areas – Angles & Phy</i> , cols. B–F
(F10)	<i>Pmetric Area Angles Exploration</i>
Cor. 3.3	<i>Lu-Fib & The Pentametric Cycle</i> , cols. AE–AJ (“Applying the quadratic equation”)
Prop. 3.4	<i>Lu-Fib & The Pentametric Cycle</i> , cols. C–L; <i>Pmetric Areas Cosine & Pi</i> , cols. O–R

4 The Doubling (Diapason) and the consonance indicator

Multiplying the area by 2 is read musically as the octave, $\delta\iota\grave{\alpha}\ \pi\alpha\sigma\acute{\omega}\nu$, the ratio 2 : 1. The last digit of $2\mathfrak{P}(n)$ is even, so the ten decimal digits collapse to the five values $\{0, 2, 4, 6, 8\}$. Halving

recovers a digit in $\{0, 1, 2, 3, 4\}$.

Definition 4.1 (Consonance indicator and musical annotation).

$$H(n) = \frac{1}{2}(2\mathfrak{P}(n) \bmod 10) = \mathfrak{P}(n) \bmod 5 \in \{0, 1, 2, 3, 4\}.$$

The musical annotation $\mathfrak{J}(n)$ is the interval assigned to $H(n)$:

H	$\mathfrak{J}$	Ratio	Greek
0	silence	—	σιγή
1	unison	1 : 1	ὁμόνυμον
2	octave	2 : 1	διὰ πασῶν
3	fifth	3 : 2	διὰ πέντε
4	fourth	4 : 3	διὰ τεσσάρων

Remark 4.2. For $h \geq 2$ the ratio is $h : (h-1)$, and $h = 1$ gives $1 : 1$. So the four non-zero values of H produce exactly the ratios formed from consecutive members of the tetractys 1, 2, 3, 4. The extraction rule ρ (Definition 2.5) is a definition. Once it is fixed, the interval at every n is computed, not chosen (§2.4). What remains interpretive is the musical meaning attached to these ratios, not the ratios themselves.

5 The fundamental period (the Pentametric Vertebra)

Proposition 5.1 (Periodicity and symmetry). *For all integers n :*

- (a) $H(n + 25) = H(n)$, and 25 is the minimal period;
- (b) $H(-n) = H(n)$; hence $H(25 - n) = H(n)$ and $H(50 - n) = H(n)$.

Proof. (a) Since $n + 25 \equiv n \pmod{5}$, the Legendre symbol is unchanged, and by (2) $\mathfrak{P}(n + 25) - \mathfrak{P}(n) = ((n + 25)^2 - n^2)/5 = 10n + 125 \equiv 0 \pmod{5}$. The minimal period divides 25, so it is 1, 5 or 25. It is not 1 or 5, because $H(1) = 1 \neq 3 = H(6)$. (b) $\left(\frac{-n}{5}\right) = \left(\frac{-1}{5}\right)\left(\frac{n}{5}\right) = \left(\frac{n}{5}\right)$, since $-1 \equiv 4$ is a square modulo 5; hence $\mathfrak{P}(-n) = \mathfrak{P}(n)$. Combining with (a) gives the reflections. $\square$

Definition 5.2 (The Pentametric Vertebra, Ὁ Πεντάμετρος Σπόνδυλος). The Vertebra is the block of $25 = 5^2$ terms $V = (H(0), H(1), \dots, H(24))$:

$$V = 0, 1, 0, 1, 4, 0, 3, 4, 2, 2, 0, 0, \mathbf{3}, \mathbf{3}, 0, 0, 2, 2, 4, 3, 0, 4, 1, 0, 1.$$

By Proposition 5.1(b), the 24 terms $H(1), \dots, H(24)$ form a palindrome. Its centre is the pair $H(12) = H(13) = 3$, the *fifth*. The leading term $H(0) = 0$ is the boundary silence that the reflection $n \mapsto 25 - n$ pairs with $H(25) = 0$. Thus the block $H(0), \dots, H(25)$ of length 26 is a palindrome; the 25-term block V by itself is not.

Proposition 5.3 (Counts in one period). *Among $H(0), \dots, H(24)$, exactly 9 terms equal 0, and each of the values 1, 2, 3, 4 occurs exactly 4 times.*

Proof. $H(n) = 0$ means $\mathfrak{P}(n) \equiv 0 \pmod{5}$, i.e. $n^2 + 4\left(\frac{n}{5}\right) \equiv 0 \pmod{25}$. If $5 \mid n$, this always holds, which gives $n \in \{0, 5, 10, 15, 20\}$. If $\left(\frac{n}{5}\right) = +1$, we need $n^2 \equiv 21 \pmod{25}$, whose solutions are $n \equiv \pm 11$. If $\left(\frac{n}{5}\right) = -1$, we need $n^2 \equiv 4 \pmod{25}$, whose solutions are $n \equiv \pm 2$. Hence the zeros are $\{0, 2, 5, 10, 11, 14, 15, 20, 23\}$, nine in total. The distribution of the non-zero values is read off from Table 4. Since H has period 25, this finite check is a complete proof. $\square$

Remark 5.4 (Anatomy of the silences). The nine silences split as $5 + 4$. Five come from the multiples of 5, where $\mathfrak{P}(5j) = 5j^2$. The other four come from the solutions of $n^2 \equiv \mp 4\left(\frac{n}{5}\right) \pmod{25}$, namely $n \equiv \pm 2, \pm 11$.

Table 4: One full period of the Pentametric Formula. $\left(\frac{n}{5}\right)$ is the Legendre symbol modulo 5.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
$\left(\frac{n}{5}\right)$	0	+	−	−	+	0	+	−	−	+	0	+	−
$\mathfrak{P}(n)$	0	1	0	1	4	5	8	9	12	17	20	25	28
$H(n)$	0	1	0	1	4	0	3	4	2	2	0	0	3

n	13	14	15	16	17	18	19	20	21	22	23	24	25
$\left(\frac{n}{5}\right)$	−	+	0	+	−	−	+	0	+	−	−	+	0
$\mathfrak{P}(n)$	33	40	45	52	57	64	73	80	89	96	105	116	125
$H(n)$	3	0	0	2	2	4	3	0	4	1	0	1	0

6 The dual structure of the Vertebra

The Vertebra admits two complementary counts.

6.1 The term count: $25 = 5^2$

Counted as terms, the Vertebra has $25 = 5^2$ elements. By Proposition 5.3, these are $9 = 3^2$ silences (Ἡ Τριαδικὴ Σιγή, the Triadic Silence) and $16 = 4^2$ consonances (Τὸ Τετράγωνον Σπέρμα, the Square Seed). The Square Seed is made of the four consonances, Αἱ Τέσσαρες Συμφωνίαι, with $4 = 2^2$ occurrences each.

6.2 The transition count: $24 = 4!$

Counted as transitions between consecutive terms, the Vertebra has 24 steps $H(n) \rightarrow H(n+1)$, $0 \leq n \leq 23$. These are the Twenty-Four Transitions, Αἱ Εἴκοσι Τέσσαρες Μεταβάσεις. Numerically, $24 = 4!$ is the number of orderings of the four consonances.

Interpretive remark. The two counts are related by $25 = 24 + 1$. The extra term is the boundary silence $H(0)$, which the reflection $n \mapsto 25 - n$ pairs with the Axis (§7). We record the coincidence $24 = 4!$ as a numerical correspondence only. The 24 transitions comprise 17 distinct ordered pairs $(H(n), H(n+1))$, several of which repeat (for instance $(0, 1)$ occurs three times). There is therefore no bijection between the transitions and the permutations of the four consonances, and we make no structural claim beyond the count.

7 The full palindrome

Proposition 7.1 (The Pentametric Palindrome, Ὁ Πενταμετρικὸς Παλάνδρομος). *The 51 terms $H(0), \dots, H(50)$ form a palindrome that mirrors at the Axis $n = 25$ (Ὁ Ἀξων), where $H(25) = 0$:*

$$H(25 + t) = H(25 - t) \quad (0 \leq t \leq 25).$$

Inside each half, the inner palindromes $H(1), \dots, H(24)$ and $H(26), \dots, H(49)$ mirror between $n = 12, 13$ and between $n = 37, 38$, at the pairs of fifths. Writing $V = (H(0), \dots, H(24))$ for the Vertebra and $\bar{V}$ for its reversal,

$$(H(0), \dots, H(50)) = V \parallel 0 \parallel \bar{V}.$$

The palindrome contains 19 silences (9 in each copy of the Vertebra, plus the Axis) and $32 = 2 \cdot 16$ consonances, 8 of each kind.

Proof. Symmetry is $H(50 - n) = H(n)$, from Proposition 5.1(b). For $26 \leq n \leq 50$, $H(n) = H(50 - n)$ with $0 \leq 50 - n \leq 24$, which is the reversal of V . The counts follow from Proposition 5.3. $\square$

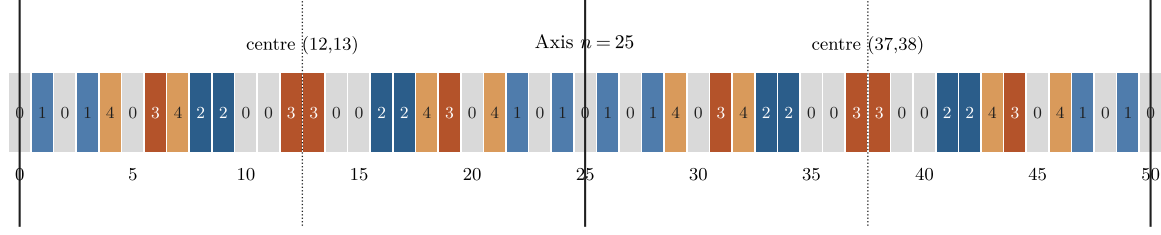


Figure 2: The sequence $H(n)$ for $n = 0, \dots, 50$. Solid lines mark the boundaries and the Axis ($n = 25$); dotted lines mark the centres of the two inner palindromes, $H(12) = H(13) = 3$ and $H(37) = H(38) = 3$ (the fifth).

Remark 7.2. Because H is even and 25-periodic, the structure repeats indefinitely. Every multiple of $25/2$ is a centre of symmetry: the Axes are at $n = 25j$, and the fifths at the centres are at $n = 25j + 12.5$. The next palindrome begins at $n = 50$, sharing its boundary silence $H(50) = 0$ with the previous one.

8 The counts and the Pythagorean triple

	One Vertebra ($n = 0, \dots, 24$)	Full Palindrome ($n = 0, \dots, 50$)
Silence ($H = 0$)	$9 = 3^2$	$19 = 2 \cdot 9 + 1$
Unison ($H = 1$)	$4 = 2^2$	8
Octave ($H = 2$)	$4 = 2^2$	8
Fifth ($H = 3$)	$4 = 2^2$	8
Fourth ($H = 4$)	$4 = 2^2$	8
Total	$25 = 5^2$	$51 = 2 \cdot 5^2 + 1$

Interpretive remark (The Pythagorean Triple, Ἡ Πυθαγόρειος Τριάς). In each Vertebra the silences and the consonances satisfy

$$3^2 + 4^2 = 5^2 : \quad 9 \text{ (silences)} + 16 \text{ (consonances)} = 25 \text{ (period)}.$$

The period $25 = 5^2$ is forced by the denominator 5 in (2). The count of silences is $5 + 4$ (Remark after Proposition 5.3). That these numbers line up with the first Pythagorean triple is a fact; whether it reflects anything deeper is left open (§15).

9 The Pentametric–Lucas Theorem

We use the classical identity [5]

$$L_k^2 - 5F_k^2 = 4(-1)^k \quad (k \geq 0). \quad (3)$$

Theorem 9.1 (The Pentametric–Lucas Theorem). *For every $k \geq 0$,*

$$\mathfrak{P}(L_k) = F_k^2, \quad \text{equivalently} \quad \sqrt{\mathfrak{P}(L_k)} = F_k.$$

Proof. Modulo 5 the Lucas numbers are periodic with period 4: $(L_0, L_1, L_2, L_3) \equiv (2, 1, 3, 4)$. Hence $\left(\frac{L_k}{5}\right) = -1, +1, -1, +1$ for $k \equiv 0, 1, 2, 3 \pmod{4}$, that is,

$$\left(\frac{L_k}{5}\right) = -(-1)^k. \quad (4)$$

In particular $5 \nmid L_k$. By (2), (4) and (3),

$$\mathfrak{P}(L_k) = \frac{L_k^2 - 4(-1)^k}{5} = \frac{5F_k^2}{5} = F_k^2. \quad \square$$

Table 5: Verification of Theorem 9.1 for $k = 0, \dots, 13$. The identity has also been checked by exact integer arithmetic for all $k \leq 59$.

k	$n = L_k$	$(\frac{n}{5})$	$\mathfrak{P}(n)$	$\sqrt{\mathfrak{P}(n)}$	Fibonacci
0	2	-1	0	0	$F_0 = 0$
1	1	+1	1	1	$F_1 = 1$
2	3	-1	1	1	$F_2 = 1$
3	4	+1	4	2	$F_3 = 2$
4	7	-1	9	3	$F_4 = 3$
5	11	+1	25	5	$F_5 = 5$
6	18	-1	64	8	$F_6 = 8$
7	29	+1	169	13	$F_7 = 13$
8	47	-1	441	21	$F_8 = 21$
9	76	+1	1156	34	$F_9 = 34$
10	123	-1	3025	55	$F_{10} = 55$
11	199	+1	7921	89	$F_{11} = 89$
12	322	-1	20736	144	$F_{12} = 144$
13	521	+1	54289	233	$F_{13} = 233$

The converse also holds: the Lucas numbers are the *only* positions at which $\mathfrak{P}$ is a perfect square.

Theorem 9.2 (Converse). *Let $n \geq 1$. Then $\mathfrak{P}(n)$ is a perfect square if and only if n is a Lucas number. In that case, $n = L_k$ and $\mathfrak{P}(n) = F_k^2$.*

Proof. Sufficiency is Theorem 9.1. Conversely, suppose $\mathfrak{P}(n) = y^2$ with $y \geq 0$. If $5 \mid n$, write $n = 5j$ with $j \geq 1$; then $y^2 = \mathfrak{P}(n) = 5j^2$, which is impossible. Otherwise $(\frac{n}{5}) = \varepsilon \in \{\pm 1\}$ and $n^2 = 5y^2 - 4\varepsilon$. If $y = 0$, then $n^2 = 4$ and $n = 2 = L_0$. If $y \geq 1$, then $5y^2 + 4$ or $5y^2 - 4$ is a square. By Gessel's criterion [4, 5], $y = F_k$ for some $k \geq 1$. By (3), $L_k^2 = 5F_k^2 + 4(-1)^k$. If $-4\varepsilon = 4(-1)^k$, then $n = L_k$. If not, then both $5F_k^2 + 4$ and $5F_k^2 - 4$ are squares. Two positive squares differing by 8 must be 9 and 1, so $F_k = 1$ and $n \in \{1, 3\} = \{L_1, L_2\}$. $\square$

Remark 9.3. Theorem 9.2 corrects an earlier informal statement of the author's: perfect squares among the values of $\mathfrak{P}$ occur *only* at Lucas positions. At every other n , $\mathfrak{P}(n)$ is an integer within $\frac{4}{5}$ of $n^2/5$ but is not a square. A direct check over $1 \leq n \leq 2 \times 10^5$ returns exactly the 26 Lucas numbers in that range.

9.1 The author's conjecture: a simple form at Lucas positions

The Geometric Lucas Meter obtains the distance to the k -th centre by a single multiplication, $d_k = L_k \cdot \sqrt{0.1}$. The author conjectured that the area F_k^2 should be obtainable from L_k in an equally simple way, without the residue machinery of the general formula. The next proposition confirms this conjecture in three forms, and shows exactly how far a single multiplication can go.

Proposition 9.4 (Simple forms at Lucas positions). *Let $d_k = L_k \sqrt{0.1}$ be the Lucas-meter distance of the k -th centre. Then:*

- (a) Multiply and correct: $F_k^2 = (L_k \sqrt{0.2})^2 - \frac{4}{5}(-1)^k = 2d_k^2 - \frac{4}{5}(-1)^k$.
- (b) Multiply and round: for every $k \geq 2$, $F_k = \text{round}(L_k \sqrt{0.2}) = \text{round}(\sqrt{2} d_k)$, and hence $F_k^2 = \text{round}(L_k \sqrt{0.2})^2$.
- (c) Neighbouring centres (exact): $F_k = \frac{L_{k-1} + L_{k+1}}{5} = \sqrt{0.4} (d_{k-1} + d_{k+1})$ for $k \geq 1$.

- (d) Limit of a pure multiplication: *there is no constant c with $F_k = c L_k$ for all k , although $F_k/L_k \rightarrow \sqrt{0.2}$.*

Proof. (a) is (3), using $(L_k \sqrt{0.2})^2 = L_k^2/5 = 2d_k^2$. (b) With $\psi = (1 - \sqrt{5})/2$, Binet's formulas $L_k = \varphi^k + \psi^k$ and $F_k = (\varphi^k - \psi^k)/\sqrt{5}$ give $L_k \sqrt{0.2} - F_k = 2\psi^k/\sqrt{5}$. For $k \geq 2$, $|2\psi^k/\sqrt{5}| \leq 2\psi^2/\sqrt{5} \approx 0.342 < \frac{1}{2}$. (c) is the classical identity $L_{k-1} + L_{k+1} = 5F_k$ [5], together with $\sqrt{0.1}\sqrt{0.4} = 0.2$. (d) $F_1/L_1 = 1$ while $F_2/L_2 = \frac{1}{3}$. The limit follows from (b). $\square$

Remark 9.5. Form (b) answers the conjecture in the sense intended. Up to rounding, the *side* of the k -th Fibonacci square is $L_k \cdot \sqrt{0.2}$, just as its distance from P is $L_k \cdot \sqrt{0.1}$. Geometrically, it is $\sqrt{2}$ times the Lucas-meter distance, i.e. the side of the square S_k of Proposition 10.3. Form (d) shows why rounding, or the exact $\mp \frac{4}{5}$ of form (a), cannot be avoided. That correction is precisely what the Pentametric Formula supplies, since $\mathfrak{P}(L_k) = (L_k \sqrt{0.2})^2 + \frac{4}{5}\chi(L_k)$ with $\chi(L_k) = -(-1)^k$. Formula (b) was verified by exact high-precision arithmetic for $2 \leq k < 120$; it fails only at $k = 0, 1$.

10 The Geometric Lucas Meter (Holden, 1975)

This section states a *classical* result. It is due to H.L. Holden [1], was developed further by Hoggatt and Alladi [2], and was rediscovered by Gardiner [3]. We claim no priority for it. Holden's paper, "Fibonacci Tiles", does not name the result. In the literature it is described in words, for example "the centres lie on orthogonal lines". In this paper we refer to it as the *Geometric Lucas Meter* (Τὸ Γεωμετρικὸν Μέτρον τοῦ Λουκά), while fully crediting Holden.

Theorem 10.1 (Holden 1975 [1]; see also [2, 3]). *Arrange the squares of sides $F_1, F_2, F_3, \dots$ in the standard Fibonacci spiral, and let C_k be the centre of the square of side F_k . Then:*

- (a) *the centres C_k with k odd lie on one straight line, and those with k even lie on a second line perpendicular to the first; the slopes are 3 and $-\frac{1}{3}$;*
- (b) *if P is the intersection of the two lines, then*

$$|C_k P| = \frac{L_k}{\sqrt{10}} = L_k \cdot \sqrt{\frac{1}{10}} \quad (k \geq 1).$$

Holden [1] proves (a) for an arbitrary generalised Fibonacci sequence f_k and gives the distances as $(f_{k+3} + f_{k-3})/(2\sqrt{10})$. For the Fibonacci numbers, $F_{k+3} + F_{k-3} = 2L_k$, which gives (b). Hoggatt and Alladi [2] rederive the centres by generating functions and state the distances as proportional to the Lucas numbers. Gardiner [3] takes the centre of the first square as origin and obtains the two lines explicitly:

$$y = 3x \quad (\text{odd } k), \quad 3y = -(x - 1) \quad (\text{even } k), \quad P = \left(\frac{1}{10}, \frac{3}{10}\right).$$

Thus $|C_1 P| = \sqrt{0.01 + 0.09} = \sqrt{0.1} = L_1/\sqrt{10}$. In other words, the Lucas number L_k counts how many units of $\sqrt{0.1} = \sqrt{10}/10$ lie between P and the centre of the k -th Fibonacci square.

Remark 10.2 (A compact form). In Gardiner's coordinates, with the spiral turning counter-clockwise as in Figure 3, put $w = -(1 + 3i)/10$, so that $|w| = 1/\sqrt{10}$. Identifying the plane with $\mathbb{C}$, the centres are

$$C_k - P = i^{k-1} L_k w \quad (k \geq 1).$$

This packages (a) and (b) together: the factor i^{k-1} alternates the centres between the two perpendicular lines, and $L_k |w|$ is the distance. We verified this form by exact rational arithmetic for $k \leq 16$.

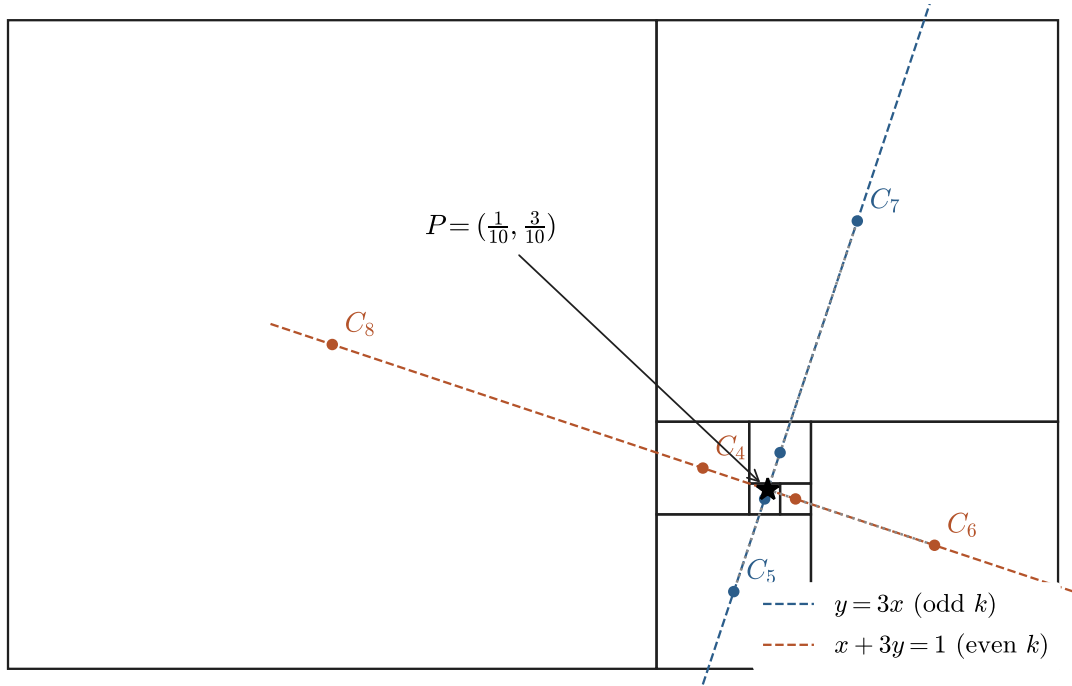


Figure 3: The Geometric Lucas Meter (Holden 1975). Fibonacci squares of sides 1, 1, 2, 3, 5, 8, 13, 21. The centres of odd-indexed squares lie on $y = 3x$ and those of even-indexed squares on $x + 3y = 1$ (Gardiner’s coordinates). The lines meet at $P = (\frac{1}{10}, \frac{3}{10})$, and $|C_k P| = L_k / \sqrt{10}$. Grey dotted segments show $|C_5 P| = 11/\sqrt{10}$, $|C_6 P| = 18/\sqrt{10}$ and $|C_7 P| = 29/\sqrt{10}$.

10.1 Correspondence with the Pentametric–Lucas Theorem

	Pentametric–Lucas Theorem (new)	Geometric Lucas Meter (Holden 1975)
Statement	$\mathfrak{P}(L_k) = F_k^2$	$ C_k P = L_k \cdot \sqrt{1/10}$
Role of L_k	argument giving the <i>area</i>	multiplier giving the <i>position</i>
Object	the Fibonacci square	the centre of that square

Together, the two statements form a complete correspondence: the Lucas number L_k simultaneously indexes the k -th Fibonacci square (via its area) and locates its centre (via its distance from P).

10.2 A geometric reading of $\mathfrak{P}$ in the Fibonacci tiling

The two statements are linked by a single geometric construction. This construction, and the identities below, are the author’s contribution. They correspond to the “ ± 0.8 ” and “ $\pm 2\pi/5$ ” constructions in the accompanying GeoGebra files.

Proposition 10.3 (Square and circle identities). *Let Q_k be the Fibonacci square of side F_k with centre C_k . Let Γ_k be the circle with centre C_k passing through P , and let S_k be the square inscribed in Γ_k with a vertex at P . Let Ω_k be the circumcircle of Q_k . Then for every $k \geq 1$:*

(a) S_k has side $L_k/\sqrt{5} = L_k\sqrt{0.2}$ and

$$\text{area}(S_k) = \frac{L_k^2}{5} = F_k^2 + \frac{4}{5}(-1)^k = \text{area}(Q_k) \pm \frac{4}{5};$$

(b) $\text{area}(\Gamma_k) - \text{area}(\Omega_k) = \frac{2\pi}{5}(-1)^k;$

(c) $\mathfrak{P}$ removes the discrepancy in (a): $\mathfrak{P}(L_k) = \text{area}(S_k) - \frac{4}{5}(-1)^k = \text{area}(Q_k)$.

Proof. By Theorem 10.1, Γ_k has radius $r_k = L_k/\sqrt{10}$. The inscribed square has diagonal $2r_k$ and side $\sqrt{2}r_k = L_k/\sqrt{5}$, so its area is $L_k^2/5$. Part (a) follows from (3). For (b), Ω_k has radius $F_k/\sqrt{2}$, so the difference of areas is $\pi(L_k^2/10 - F_k^2/2) = \frac{\pi}{10}(L_k^2 - 5F_k^2) = \frac{2\pi}{5}(-1)^k$. Part (c) is (2) combined with (4). $\square$

$$k=4: F_4^2=9, L_4^2/5=49/5=F_4^2+\frac{4}{5}$$

$$k=5: F_5^2=25, L_5^2/5=121/5=F_5^2-\frac{4}{5}$$

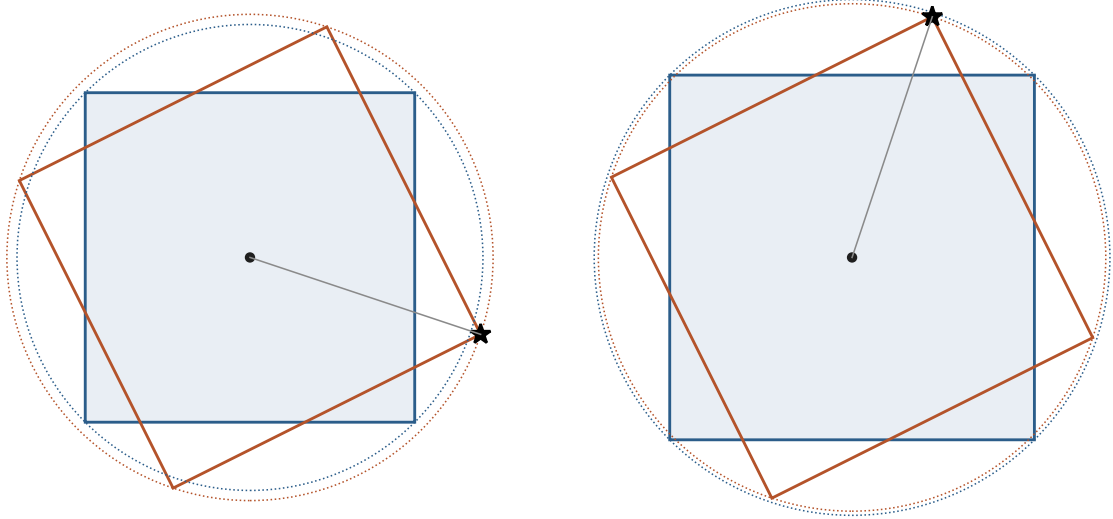


Figure 4: Proposition 10.3 for $k=4$ and $k=5$. Shaded: the Fibonacci square Q_k . Red: the square S_k centred at C_k with a vertex at P (star); its area is $L_k^2/5 = F_k^2 \pm \frac{4}{5}$. Dotted: the circles Γ_k (through P) and Ω_k (circumcircle of Q_k), whose areas differ by $\pm \frac{2\pi}{5}$.

10.3 The angles of the construction

The author's angle tables (**Angles.xlsx**) tabulate arctan of the ratios $1/3$, $1/2$, $3/4$, $4/3$ and others, together with their sines, cosines and doubles. Two of these angles govern the geometry above.

Proposition 10.4 (The two angles). *Let $\theta_1 = \arctan \frac{1}{3}$ and $\theta_2 = \arctan \frac{1}{2}$. Then:*

- (a) $\sin \theta_1 = \sqrt{0.1}$ and $\sin \theta_2 = \sqrt{0.2}$. These are the unit of the Geometric Lucas Meter and the side unit $n\sqrt{0.2}$ of Proposition 3.4.
- (b) Holden's lines $y = 3x$ and $x + 3y = 1$ make the angle θ_1 with the vertical and horizontal axes of the tiling, respectively. The vector from C_1 to P is $\sqrt{0.1}(\sin \theta_1, \cos \theta_1)$.
- (c) For every $k \geq 1$, the sides of the square S_k of Proposition 10.3 have slopes $\frac{1}{2}$ and -2 , that is, they are inclined at θ_2 to the axes of the tiling.
- (d) $\theta_1 + \theta_2 = \frac{\pi}{4}$, and $\tan 2\theta_1 = \frac{3}{4}$, $\tan 2\theta_2 = \frac{4}{3}$. Thus $2\theta_1$ and $2\theta_2$ are the two acute angles of the 3–4–5 right triangle, and $2\theta_1 + 2\theta_2 = \frac{\pi}{2}$.

Proof. (a) $\sin \arctan t = t/\sqrt{1+t^2}$, which gives $\frac{1/3}{\sqrt{10/9}} = \frac{1}{\sqrt{10}}$ and $\frac{1/2}{\sqrt{5/2}} = \frac{1}{\sqrt{5}}$. (b) This is immediate from the slopes 3 and $-\frac{1}{3}$ and from $P - C_1 = (\frac{1}{10}, \frac{3}{10})$. (c) In the notation of the remark following Theorem 10.1, $P - C_k = -i^{k-1}L_k w$, whose direction is $\arg(1+3i) + (k-1)\frac{\pi}{2} = \frac{\pi}{2} - \theta_1 + (k-1)\frac{\pi}{2}$. A side of the square inscribed in Γ_k with a vertex at P is

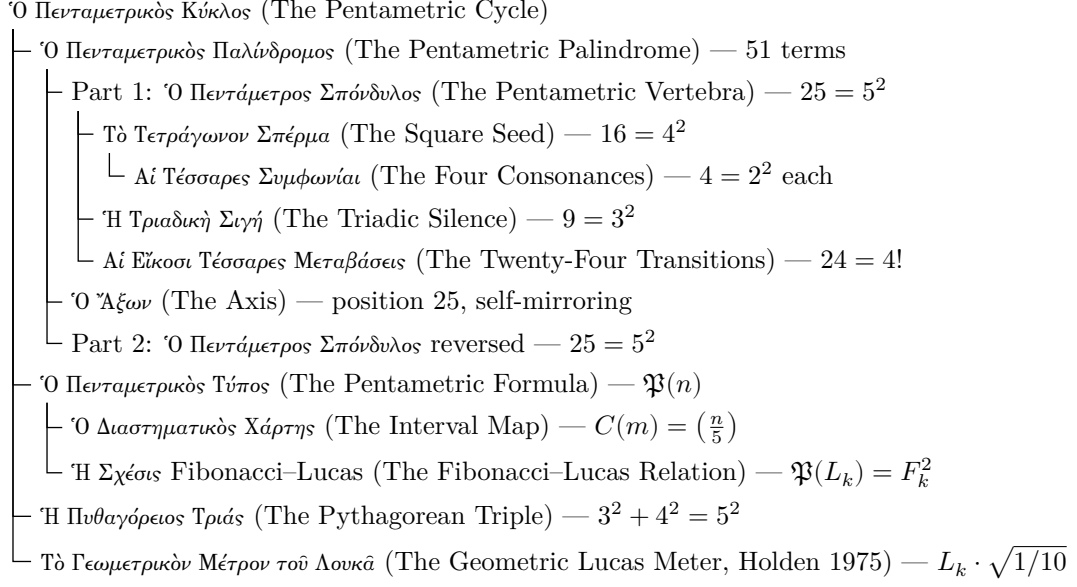
$(i-1)(P-C_k)$, which turns this direction by $\frac{3\pi}{4}$. Modulo $\frac{\pi}{2}$, the side direction is therefore $\frac{\pi}{4} - \theta_1 = \theta_2$. (d) $\tan(\theta_1 + \theta_2) = \frac{1/3+1/2}{1-1/6} = 1$ (Euler's arctangent identity). $\tan 2\theta_1 = \frac{2/3}{1-1/9} = \frac{3}{4}$ and $\tan 2\theta_2 = \frac{1}{1-1/4} = \frac{4}{3}$. $\square$

Interpretive remark. Proposition 10.4(d) is a *geometric* appearance of the triple $3^2 + 4^2 = 5^2$ in the Fibonacci tiling. It is independent of the count $9 + 16 = 25$ of §8. The orientation of Holden's lines (θ_1) and the orientation of the squares S_k (θ_2) are the half-angles of the 3–4–5 triangle.

11 The complete nomenclature

Greek	Transliteration	English	Role
Ὁ Πενταμετρικὸς Κύκλος	Ho Pentametricos Kyklos	The Pentametric Cycle	Whole system
Ὁ Πενταμετρικὸς Παλίνδρομος	Ho Pentametricos Palindromos	The Pentametric Palindrome	Full palindrome (51)
Ὁ Πεντάμετρος Σπόνδυλος	Ho Pentametros Spondylos	The Pentametric Vertebra	Fundamental period (25)
Αἱ Εἴκοσι Τέσσαρες Μεταβάσεις	Hai Eikosi Tessares Metabaseis	The Twenty-Four Transitions	Transitions (24)
Ὁ Ἀξὼν	Ho Axōn	The Axis	Mirror point ($n = 25$)
Τὸ Τετράγωνον Σπέρμα	To Tetragōnon Sperma	The Square Seed	Consonance skeleton (16)
Ἡ Τριαδικὴ Σιγή	Hē Triadikē Sigē	The Triadic Silence	Silence skeleton (9)
Αἱ Τέσσαρες Συμφωνίαι	Hai Tessares Symphōniai	The Four Consonances	The intervals (4)
Ἡ Πυθαγόρειος Τριάς	Hē Pythagoreios Trias	The Pythagorean Triple	Structural identity
Ὁ Πενταμετρικὸς Τύπος	Ho Pentametricos Typos	The Pentametric Formula	Generator ($\mathfrak{P}$)
Ὁ Διαστηματικὸς Χάρτης	Ho Diastēmatikos Chartēs	The Interval Map	Ratio mapper $C(m)$
Ὁ Ἀρμονικὸς Κανὼν	Ho Harmonikos Kanōn	The Harmonic Canon	Alternative name of $C(m)$
Τὸ Γεωμετρικὸν Μέτρον τοῦ Λουκά	To Geōmetrikon Metron tou Louka	The Geometric Lucas Meter	Distance to square centres (Holden 1975)

12 The structure diagram



13 Musical significance

This section separates two kinds of statement. The first subsection lists facts that follow by proof once the ratio rule $\rho(H) = H : (H - 1)$ of Definition 2.5 is fixed; they are computed, not interpreted. The second lists readings that connect the cycle to the Pythagorean tradition; they are interpretive and are not part of the mathematical claims.

13.1 Derived musical facts

1. *All four consonances, in equal number.* The ratios $\rho(H(n))$ take exactly the values $1 : 1$, $2 : 1$, $3 : 2$ and $4 : 3$, each four times per Vertebra and eight times per Palindrome, with nine (respectively nineteen) silences (Propositions 5.3 and 7.1).
2. *The consonances are the tetractys ratios.* For $H = 2, 3, 4$ the single rule $H : (H - 1)$ yields $2 : 1$, $3 : 2$, $4 : 3$: the octave, fifth and fourth, i.e. exactly the ratios of consecutive members of 1, 2, 3, 4 [7, 8].
3. *The fifth at the centre.* The two inner palindromes are centred on the pairs $H(12) = H(13) = 3$ and $H(37) = H(38) = 3$, so the interval at each centre is the fifth $3 : 2$ (Proposition 5.1).
4. *Symmetric distribution.* The interval sequence $n \mapsto \rho(H(n))$ is 25-periodic and mirrors at every Axis $n = 25j$ (Propositions 5.1 and 7.1).
5. *A computed annotation.* The musical annotation $\mathfrak{J}(n) = \rho(H(n))$ is determined at every position n by the seven-step chain of §2.4.

13.2 Interpretive readings

6. The doubling $2\mathfrak{P}$ is read as the diapason (octave), the fundamental musical operation; halving then recovers the consonance indicator.
7. In the Pythagorean tradition, the fifth is the generator of the scale: Pythagorean tuning is built by stacking fifths and reducing by octaves [6]. That the fifth occupies the centre of the palindrome (item 3) is therefore suggestive.

8. The period splits as $3^2 + 4^2 = 5^2$ (silences, consonances, period), recalling the first Pythagorean triple; the same triple appears geometrically in the angles of the construction (Proposition 10.4).
9. The 24 transitions of a Vertebra match in number, though not one-to-one (§6), the $4!$ orderings of the four consonances.

14 Mathematical significance

1. The Pentametric Formula has the closed form $\mathfrak{P}(n) = (n^2 + 4(\frac{n}{5}))/5$. It is the integer obtained by shifting $n^2/5$ by $\pm\frac{4}{5}$ according to the quadratic character of n modulo 5 (Lemma 2.3).
2. The ten forms (F1)–(F10) of $\mathfrak{P}$ (Theorem 3.2) reach the same areas through digits, floors, a quintic polynomial, a Gauss sum, golden-ratio rotations and pentagonal angles. They are unified by the quadratic character modulo 5.
3. The cycle is nested: Palindrome (51) $\supset$ Vertebra (25) $\supset$ Square Seed (16) $\supset$ each Consonance (4). The counts 25, 16, 9, 4 are $5^2, 4^2, 3^2, 2^2$.
4. The consonance sequence H is even and has minimal period 25, so it is palindromic about every multiple of $25/2$ (Proposition 5.1).
5. The Pentametric–Lucas Theorem $\mathfrak{P}(L_k) = F_k^2$ holds for all $k \geq 0$, and conversely the perfect-square values of $\mathfrak{P}$ occur exactly at the Lucas numbers (Theorems 9.1–9.2). At Lucas positions the side is $\text{round}(L_k\sqrt{0.2})$, mirroring the distance $L_k\sqrt{0.1}$ (Proposition 9.4).
6. In the Fibonacci tiling, the Lucas number L_k both returns the area of the k -th square through $\mathfrak{P}$ and measures the distance $L_k\sqrt{1/10}$ to its centre (Holden 1975). The square and circle identities of Proposition 10.3 make the correction term $\pm\frac{4}{5}$ of $\mathfrak{P}$ visible as a geometric area.
7. To the best of our knowledge, neither the sequence $\mathfrak{P}(n)$ (0, 1, 0, 1, 4, 5, 8, 9, 12, 17, ...) nor the sequence $H(n)$ appears in the On-Line Encyclopedia of Integer Sequences [11] (searched October 2026).

15 Discussion

The Pentametric Cycle sits at the intersection of elementary number theory, the geometry of the Fibonacci tiling, and the Pythagorean theory of consonance. Its connections to classical results are the following.

- The Fibonacci–Lucas identity $F_k^2 = (L_k^2 - 4(-1)^k)/5$ is the identity underlying Theorem 9.1. The Pentametric Formula extends its right-hand side from the Lucas numbers to all integers by replacing $-(-1)^k$ with the Legendre symbol $(\frac{n}{5})$, which agrees with it on Lucas numbers by (4).
- Gessel’s test for Fibonacci numbers [4] is equivalent to the statement that the square values of $\mathfrak{P}$ occur exactly at the Lucas numbers (Theorem 9.2).
- Holden [1] showed that the centres of spiralling Fibonacci squares lie on orthogonal lines at distances proportional to the Lucas numbers. This geometric result complements the Pentametric–Lucas Theorem: the Lucas number simultaneously indexes the square (via area) and locates its centre (via distance). Proposition 10.3 shows that the two are linked by a single construction.

- The cosine and golden-rotation forms (F8)–(F9) are instances of the quadratic Gauss sum modulo 5, $\cos \frac{2\pi n}{5} - \cos \frac{4\pi n}{5} = \frac{\sqrt{5}}{2} \left(\frac{n}{5}\right)$ [10]. This explains why the angles 72° and 144° of the regular pentagon appear in the author’s constructions.
- Holden’s lines are inclined at $\arctan \frac{1}{3}$, and the squares S_k at $\arctan \frac{1}{2}$. Their doubles are the acute angles of the 3–4–5 triangle (Proposition 10.4).
- The doubling $2\mathfrak{P}$ is the musical octave, and the halving recovers the consonances. The split $9 + 16 = 25$ recalls the triple $3^2 + 4^2 = 5^2$.

Open questions.

1. *Other moduli.* Replacing 5 by another prime p gives the analogous construction $\mathfrak{P}_p(n) = (n^2 + c(n/p))/p$. For which p and c is it integer-valued? Does the analogous consonance sequence have period p^2 , and does its zero count again split as a sum of two squares?
2. *Other recurrences.* Holden’s lines exist for every generalised Fibonacci sequence [1, 2]. Is there a Pentametric formula whose square values are exactly f_k^2 at the companion sequence for a generalised Fibonacci sequence f_k , or for Pell and Pell–Lucas numbers?
3. *Other triples.* Is there a companion cycle whose counts realise another Pythagorean triple, such as $5^2 + 12^2 = 13^2$?
4. *Continued fractions.* $\mathfrak{P}(n)$ is determined by $n^2/5$ and the quadratic character of n modulo 5. What is the precise connection to the continued fraction expansion of $\sqrt{5}$, and to rational approximations of φ ?
5. *The transitions.* Does the multiset of the 24 transitions, with its 17 distinct pairs, have an intrinsic combinatorial description?

16 Conclusion

The Pentametric Cycle is a family of equivalent formulas generating a nested palindrome that encodes the four Pythagorean consonances. We have shown that the Pentametric Formula equals $(n^2 + 4(\frac{n}{5}))/5$, that its consonance sequence has period 25 with the counts $9 + 16 = 25$, and that it satisfies the Pentametric–Lucas Theorem $\mathfrak{P}(L_k) = F_k^2$ together with its converse. Holden’s classical result of 1975 on Fibonacci tiles provides a complementary geometric perspective, in which the Lucas number simultaneously indexes the square and locates its centre. The square and circle identities of Proposition 10.3 join the two perspectives. The formula and its nomenclature offer a new perspective on the Pythagorean tradition, and open several directions for further research.

Acknowledgements

The geometric result on the centres of Fibonacci squares is due to H.L. Holden (1975); the connection to the Pentametric Cycle, the Pentametric–Lucas Theorem and its converse, and the square and circle identities of Proposition 10.3 are the author’s contribution.

The author used an AI assistant (Claude, developed by Anthropic) to help write the manuscript and to check the proofs and numerical results independently. The author is responsible for the content of the paper.

Data and materials availability

The GeoGebra constructions, spreadsheets and images that accompany this paper are available in the author's repository *The Pentametric Cycle*, <https://github.com/lluisgarcia/The-Pentametric-Cycle>. All numerical statements in this paper were verified independently by exact integer and rational arithmetic.

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A Computational verification

All claims were checked with exact arithmetic (Python `fractions`):

- Definition 2.1 and the closed form (2) agree, and $\mathfrak{P}(n) \in \mathbb{Z}$, for $0 \leq n < 10^4$; the residue m takes only the values 0, 4, 6.
- H has minimal period 25; the blocks $H(1..24)$, $H(0..25)$ and $H(0..50)$ are palindromes, while $H(0..24)$ is not.
- Counts: $\{0:9; 1,2,3,4:4\}$ on $0..24$; $\{0:19; 1,2,3,4:8\}$ on $0..50$.
- $\mathfrak{P}(L_k) = F_k^2$ for $0 \leq k \leq 59$; for $1 \leq n \leq 2 \times 10^5$, $\mathfrak{P}(n)$ is a square exactly at the 26 Lucas numbers in that range.

- The ten forms (F1)–(F10) of Theorem 3.2, the side-length correction of Corollary 3.3 and both defects of Proposition 3.4 agree with $\mathfrak{P}(n)$ for $0 \leq n < 3000$ (exactly for the rational forms, to 10^{-6} relative error for the trigonometric ones). This was checked independently of, and in agreement with, the author’s spreadsheets.
- The Fibonacci spiral was built with exact rational coordinates for $k \leq 16$. The odd and even centres are collinear on $y = 3x$ and $x + 3y = 1$; $10|C_k P|^2 = L_k^2$; and $C_k - P = i^{k-1} L_k w$.